

Lecture Notes 3: Linear Algebra

Our attention for this section/chapter will be to understand linear systems of a different kind. We've seen ones that look like:

$$3x - 4y = 5$$

$$2x - 3y = 1$$

And there are systems like

$$\frac{dx}{dt} = 3x - 4y$$

$$\frac{dy}{dt} = 2x - 3y$$

Both can be written more shortly as

$$\begin{pmatrix} 3 & -4 \\ 2 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 3 & -4 \\ 2 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix}$$

depending on the system

Even more succinctly: $A \underline{x} = \underline{b}$ or $\underline{\dot{x}} = A \underline{x}$
depending on the system.

Thus it is important to understand what information we can glean from these systems.

Basic Definitions

A matrix is a rectangular array of elements (or entries) arranged in rows (horizontal) and columns (vertical):

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} \quad a_{(\text{row}, \text{column})}$$

Matrices are always written w/ capital letters.

The above matrix is an $m \times n$ matrix ("m by n")

Sometimes we write $A = [a_{ij}]$ to indicate the matrix, if the size is known

If $m=n$, then A is a square matrix.

If the only entries of a square matrix that are non-zero are ones where $i=j$, we say the matrix is diagonal.

$$D = \begin{pmatrix} d_{11} & & \\ & d_{22} & \\ & & \ddots \\ & & & d_{nn} \end{pmatrix}$$

we usually don't bother writing so many zeros, it's a waste of effort.
[MATLAB: `sparse(m, n)`]

If all the diagonal elements of a diagonal matrix are equal to 1, we call that the $n \times n$ identity matrix.

Notation: I_n , $\mathbb{1}_n$, $\text{Id}(n)$, etc...

$$I_n = \begin{pmatrix} 1 & & \\ & 1 & \\ & & \ddots \\ & & & 1 \end{pmatrix}$$

Blank spaces are zero.

A $m \times 1$ matrix is called a column vector: $\begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix}$

A $1 \times n$ matrix is called a row vector: $(x_1, \dots, x_n)$

Thus, a $m \times n$ matrix can be viewed as a collection of n column vectors, or m row vectors:

$$\begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{pmatrix} \begin{pmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{pmatrix} \dots \begin{pmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{pmatrix} \quad \begin{array}{l} n \text{ column vectors} \\ \text{together form a } (m \times n)\text{-matrix} \end{array}$$

$$\begin{array}{c} (a_{11}, a_{12}, \dots, a_{1n}) \\ (a_{21}, a_{22}, \dots, a_{2n}) \\ \vdots \\ (a_{m1}, a_{m2}, \dots, a_{mn}) \end{array} \quad \begin{array}{l} m \text{ row vectors} \\ \text{together form a } (m \times n)\text{-matrix} \end{array}$$

If a matrix has all zeros for entries, it is called the zero matrix. O_{mn} or O
Two matrices are equal iff their entries all equal each other.

Ex/ $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ is a 2×2 matrix

Ex/ $\begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}$ is a diagonal matrix

Ex/ $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$ is a matrix of size $(_, _)$ ↙?

Ex/ $[2, 1, 3]$ is a vector of size $(_, _)$?

Ex/ $\begin{pmatrix} 2 \\ 0 \\ 9 \end{pmatrix}$ is a vector of size $(_, _)$?

Matrix Arithmetic

Matrices add and subtract component-wise:

$$A \pm B = [a_{ij}] \pm [b_{ij}] = [a_{ij} \pm b_{ij}]$$

Note! If dimensions of matrices do not correspond then we cannot add nor subtract them! That operation is as of yet undefined.

$$\text{Ex/ } [2, 0, 9] + [8, 1, 8] = [2+8, 0+1, 9+8] = [10, 1, 17]$$

$$\text{Ex/ } \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} - \begin{bmatrix} 6 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1-6 & 2-5 & 3-4 \\ 4-3 & 5-2 & 6-1 \end{bmatrix} = \begin{bmatrix} -5 & -3 & -1 \\ 1 & 3 & 5 \end{bmatrix}$$

We multiply a constant (or scalar) to a matrix by multiplying all entries by that scalar:

$$kA = k[a_{ij}] = [k a_{ij}] = [a_{ij} k] = Ak$$

$$\text{Ex/ } 3[2, 0, 9] = [3 \cdot 2, 3 \cdot 0, 3 \cdot 9] = [6, 0, 27] = [2, 0, 9] \cdot 3$$

$$\text{Ex/ } 4 \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = \begin{bmatrix} 4 & 8 & 12 \\ 16 & 20 & 24 \end{bmatrix}$$

$$\text{Ex/ } 2i \begin{pmatrix} 2i \\ 3 \\ -4 \end{pmatrix} = \begin{pmatrix} -4 \\ 6i \\ -8i \end{pmatrix}$$

Important!!

Properties of Matrix Addition & Scalar Multiplication

If $A, B,$ and C are $m \times n$ matrices, c & k are scalars, then the following are true:

① $A+B$ is $m \times n$

Closure

② $A+B = B+A$

Additive Commutativity

③ $A+(B+C) = (A+B)+C$

Additive Associativity

④ $c(kA) = (ck)A$

Scalar Multiplicative Associativity

⑤ $A+O = A$

Zero Elements

⑥ $A+(-A) = O$, $-A = (-1)A$

Additive Inverse

⑦ $c(A+B) = cA + cB$

Left Distributivity of Scalar Multiplication

⑧ $(c+k)A = cA + kA$

Right Distributivity of Scalar Addition

⑨ $cA = Ac$

Commutativity of Scalar Multiplication

Can prove all of these using the definitions, and properties of the matrix entries.
(see Ex. 5 on p. 119 for examples of how to prove).

The Scalar Product

This is going to be the foundation of all matrix-matrix interactions.

Def'n: The scalar product of a row vector $\vec{x}$ w/ a column vector $\vec{y}$ is given by

$$\vec{x} \cdot \vec{y} = [x_1, x_2, \dots, x_n] \cdot \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

$$= x_1 y_1 + x_2 y_2 + \dots + x_n y_n$$

The scalar product always produces a scalar.

$$\text{Ex} // [8, 5, 1] \cdot \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} = (8)(1) + (5)(-1) + (1)(2) = 8 - 5 + 2 = 5 \quad \checkmark$$

$$\text{Ex} // [3, 2] \cdot \begin{bmatrix} -4 \\ 6 \end{bmatrix} = (3)(-4) + (2)(6) = 0$$

Def'n: If two vectors have a zero scalar product, we say they are orthogonal to each other.

Def'n: (Absolute Value) For any vector $\vec{v} \in \mathbb{R}^n$, the length or absolute value of $\vec{v}$ is a nonnegative scalar, defined by:

$$\|\vec{v}\| = (\vec{v} \cdot \vec{v})^{1/2}$$

Sometimes we will call it a norm.

Vectors w/ length 1 are called unit vectors.

Matrix Multiplication

Defn: (Matrix Product) The matrix product of an $m \times r$ matrix A and a $r \times n$ matrix B is a $m \times n$ matrix C given by

$$C = AB$$

Formally: $C_{ij} = [a_{i1}, a_{i2}, \dots, a_{ir}] \begin{bmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{rj} \end{bmatrix}$

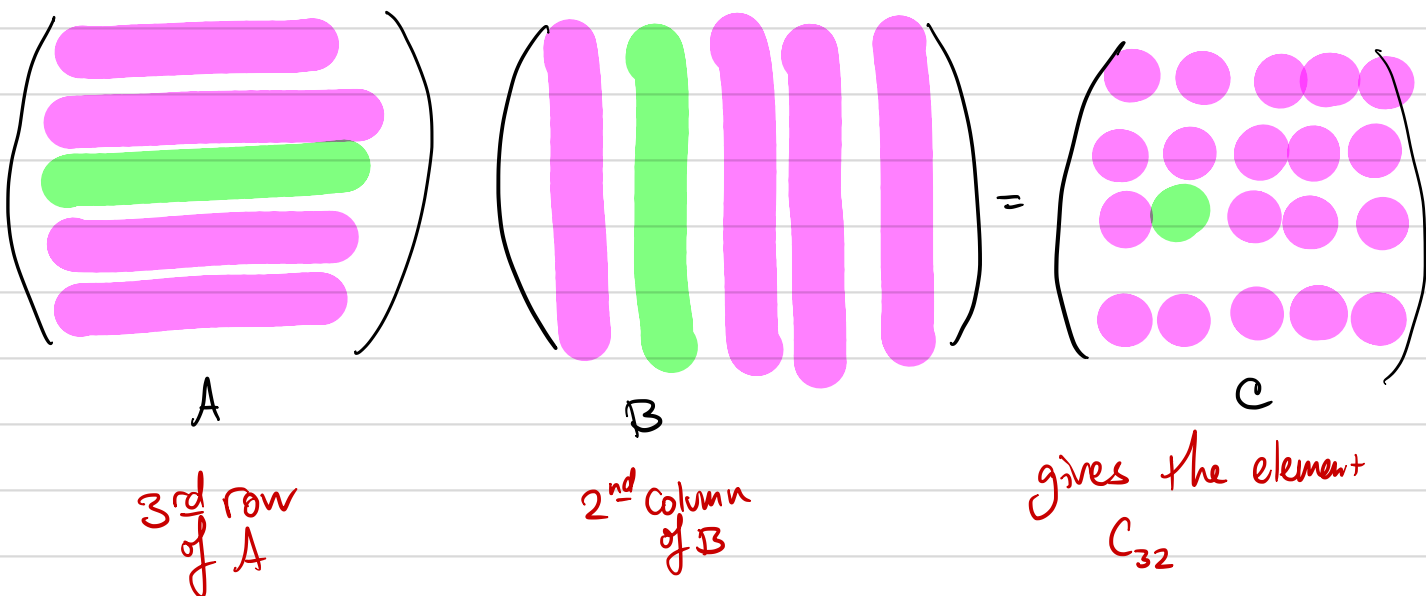
or

$$C_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{ir}b_{rj} = \sum_{k=1}^r a_{ik}b_{kj}$$

also: $C_{ij} = a_{ik}b_{kj} \leftarrow$ "repeated indices adjacent to each other summed,"
the so-called Einstein Summation Convention.

This is how one computes the matrix products, element-by-element.

Schematically: You take the i^{th} row of A , the j^{th} column of B , compute their scalar product, to get the C_{ij} element of C .



$$\text{Ex// } A = \begin{bmatrix} 1 & -1 & 3 \\ 0 & 4 & 2 \end{bmatrix} \quad B = \begin{bmatrix} 3 & 1 \\ 2 & -4 \\ -1 & 0 \end{bmatrix}$$

$2 \times 3 \qquad 3 \times 2$

Inner numbers match! Result will be: ? $\times$?

$$\begin{pmatrix} 1 & -1 & 3 \\ 0 & 4 & 2 \end{pmatrix} \begin{pmatrix} 3 & 1 \\ 2 & -4 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} (1)(3) + (-1)(2) + (3)(-1) & (1)(1) + (-1)(-4) + (3)(0) \\ (0)(3) + (4)(2) + (2)(-1) & (0)(1) + (4)(-4) + (2)(0) \end{pmatrix}$$

← 4 scalar products to compute 2×2 elements

$$= \begin{pmatrix} -2 & 5 \\ 6 & -16 \end{pmatrix}$$

$$\text{Ex// } \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 2 & 0 \end{pmatrix} = \begin{pmatrix} 5 & -1 \\ 11 & -3 \end{pmatrix}$$

$$\text{Ex// } \begin{pmatrix} 1 & -1 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} -2 & -2 \\ 2 & 4 \end{pmatrix}$$

Conclusion: In general $AB \neq BA$! Matrix products are not commutative!

$$\text{Ex// } \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

$$\text{Ex// } \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

Conclusion: $IA = AI$

In general, the Identity matrix commutes w/ all matrices w/ whom the product makes sense.

$$\text{Ex// } \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = \begin{pmatrix} a & b & c \\ 2d & 2e & 2f \\ g & h & i \end{pmatrix}$$

Conclusion: ??

$$\text{Ex// } \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} a & 2b & c \\ d & 2e & f \\ g & 2h & i \end{pmatrix}$$

Conclusion: ??

$$\text{Ex} // \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^2 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix}$$

$$\text{Ex} // \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^3 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 2 \\ 2 & 1 \end{pmatrix}$$

$$\text{Ex} // \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^4 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 3 & 2 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 5 & 3 \\ 3 & 2 \end{pmatrix}$$

Properties of Matrix Multiplication

- ① $(AB)C = A(BC)$ Associativity
- ② $A(B+C) = AB+AC$ Left Distributive
- ③ $(B+C)A = BA+CA$ Right Distributive
- ④ $AB \neq BA$ except in special cases Noncommutativity

If A is an $m \times n$ matrix, then

$$\left. \begin{array}{ll} \text{⑤ } A I_n = A & \text{⑦ } A O_{np} = O_{mp} \\ \text{⑥ } I_m A = A & \text{⑧ } O_{pm} A = O_{pn} \end{array} \right\} \text{ for any } p \neq q$$

The Matrix Transpose

It turns out that many operations can be simplified and better understood if we define something called a transpose. This is an operation where one swaps the row & column of a given entry.

If $A = [a_{ij}]$, then its transpose is written $A^T = [a_{ij}]^T = [a_{ji}]$

$$\text{Ex} // \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}^T = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix} \quad \text{Notice entries along the main diagonal remain unchanged.}$$

MATLAB: A'

$\downarrow 2 \times 3$
 $\downarrow 3 \times 2$
 ex// $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$

ex// $[2, 0, 9]^T = \begin{bmatrix} 2 \\ 0 \\ 9 \end{bmatrix}$

ex// Thus, if $\vec{v}$ is a column vector, $\|\vec{v}\|^2 = \vec{v}^T \vec{v}$

ex// If $\vec{a}$ is a row vector, $\|\vec{a}\|^2 = \vec{a} \vec{a}^T$

Properties of Transposes

① $(A^T)^T = A$ ② $(A+B)^T = A^T + B^T$

③ $(kA)^T = k(A^T)$ for any scalar k

④ $(AB)^T = B^T A^T$

prove these w/o looking them up!

A matrix is called symmetric if $A = A^T$. It is antisymmetric or skew-symmetric if $A^T = -A$

ex// $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}^T = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$. This is a symmetric matrix.

ex// $\begin{pmatrix} 0 & -\beta \\ \beta & 0 \end{pmatrix}^T = \begin{pmatrix} 0 & \beta \\ -\beta & 0 \end{pmatrix} = -\begin{pmatrix} 0 & \beta \\ \beta & 0 \end{pmatrix}$ is skew-symmetric.

// If a matrix has functions for entries, they behave [almost] like what you might expect.

① $[A(t) + B(t)]' = A'(t) + B'(t)$; where the prime indicates a derivative w.r.t. t , and is applied to each entry.

② $[cA(t)]' = cA'(t)$

③ $[A(t)B(t)]' = A'(t)B(t) + A(t)B'(t)$ [mult. order matters!]

Trace of a Matrix

The trace of a matrix is the sum of the diagonal entries of a square matrix.

$$\text{Ex/ } \text{Tr} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = 1 + 4 = 5$$

$$\text{Ex/ } \text{Tr} \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} = 1 + 5 + 9 = 15$$

It will be shown later that the Trace is a very geometrically significant object, as well as algebraically useful!

$$\text{Ex/ } \text{Tr}(A) = \text{Tr} \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & & a_{nn} \end{pmatrix} = \sum_{i=1}^n a_{ii} = a_{11} + a_{22} + \dots + a_{nn}$$

Ex/ Easy to prove that $\text{Tr}(A) = \text{Tr}(A^T)$

$$\text{LHS: } \text{Tr}(A) = \text{Tr}([a_{ij}]) = \sum a_{kk} \quad \checkmark$$

$$\text{RHS: } \text{Tr}(A^T) = \text{Tr}([a_{ji}]) = \sum a_{kk} \quad \checkmark$$

This was #49.

// An example, how to deal w/ some matrix problems.

#67) The square root of a matrix A is a matrix R s.t. $R^2 = A$. Show that the following matrix has no square root:

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$\text{Suppose } R = \begin{pmatrix} a & b \\ c & d \end{pmatrix}. \text{ Then } \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

This gives four equations!

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

← arranged by colors for convenience.

Thus $a^2 + bc = 0$
 $ca + dc = 0$

$ab + bd = 1$
 $cb + d^2 = 0$

Factoring will clarify much!

$a^2 + bc = 0$
 $c(a+d) = 0$

$b(a+d) = 1$
 $d^2 + bc = 0$

Both the green and blue equations have $a+d$ in them. One is zero, the other isn't. This means $a+d \neq 0$, which means

$$\boxed{c = 0}$$

If $c = 0$, then examining the red equation reveals

$$a^2 + b(0) = 0 \Rightarrow a^2 = 0 \text{ or } \boxed{a = 0}$$

likewise w/ the orange equation: $\boxed{d = 0}$

Finally, looking at the blue equation: $b(a+d) = 1$ or $b(0+0) = 1$, and this is impossible.

Therefore $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ does not have a square root.